

Math2010 Advanced Calculus I, 2025-2026

All Questions, By Category

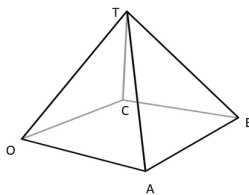
June 17, 2026

This is a collection of every problem in the course, organised by category. All problems here exist on an assignment (A), tutorial sheet (T), midterm exam (M), or practice problem set (P), as indicated by the code. Feel free to use this sheet to check that you are familiar with the various types of questions which may be asked in the final exam.

Euclidean Geometry

A1Q1. Let $\vec{u} = (2, -2, 1)$, $\vec{v} = (0, 2, 3)$ and $\vec{w} = (-5, 6, -1)$ be vectors in $\mathbb{R}^3$. Show that $\{\vec{u}, \vec{v}, \vec{w}\}$ are linearly dependent.

A1Q2. Consider the following square-based pyramid in $\mathbb{R}^3$. The vertices of the square base are $O = (0, 0, 0)$, $A = (l, 0, 0)$, $B = (l, l, 0)$, $C = (0, l, 0)$, and the tip vertex $T = (a, b, h)$ lies directly above the interior of the square base $OABC$.



- Calculate the cross-product $\vec{AB} \times \vec{AT}$ and the triple product $(\vec{AB} \times \vec{AT}) \cdot \vec{AO}$.
- Show that the area of the triangles $\triangle OAT$ and $\triangle ABT$ are equal if and only if $a + b = l$.
- Show that all four triangles have equal area if and only if $a = b = \frac{l}{2}$.

A1Q3. Let $\vec{x}, \vec{y} \in \mathbb{R}^n$.

- Show that either $(\vec{x} + \vec{y}) \cdot \vec{x} \geq 0$, or $(\vec{x} + \vec{y}) \cdot \vec{y} \geq 0$.
- Assume now that $\vec{x}$ and $\vec{y}$ are non-zero vectors of the same length. Show that, if the vector $\vec{x} + \vec{y}$ is non-zero, then it bisects the angle between $\vec{x}$ and $\vec{y}$.

A1Q4. Define the planes P_1 , P_2 and P_3 in $\mathbb{R}^3$ as follows:

- P_1 is the plane through the points $(1, 0, 0)$, $(0, 1, 0)$ and $(1, 0, 1)$,
- P_2 is the plane through $(1, 0, 0)$ with normal $(0, 0, 1)$,
- P_3 is the plane through $(-1, 2, 0)$ with tangent vectors $\vec{u} = (0, 0, 1)$, $\vec{v} = (-1, 1, 0)$.

- Express P_1 , P_2 and P_3 in parametric form.
- Show that two of these three planes are the same. (*Hint: to show two planes are the same, you can show that there is a point A in both, and a shared normal $\vec{n}$*)
- The intersection of the three planes $P_1 \cap P_2 \cap P_3$ is a line in $\mathbb{R}^3$. Describe this line using two equations in the variables x, y, z .

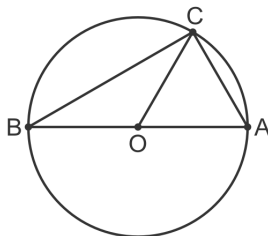
T1Q1. Consider the 4-dimensional vectors $\mathbf{a} = (-1, 3, 2, 1)$ and $\mathbf{b} = (3, 1, -2, -1)$. Calculate:

- a) $\|\mathbf{a}\|, \|\mathbf{b}\|$,
- b) $\mathbf{a} + \mathbf{b}$,
- c) $\mathbf{a} \cdot \mathbf{b}$,
- d) The angle θ between $\mathbf{a}$ and $\mathbf{b}$.

Verify that the Cauchy-Schwarz inequality and the triangle inequality hold for these two vectors.

T1Q2. Consider the following circle based at O , with diameter AB .

Define the vectors $\vec{a} := \overrightarrow{OA}$, $\vec{b} := \overrightarrow{OB}$, $\vec{c} := \overrightarrow{OC}$.



- a) Write the vectors $\overrightarrow{AB}$, $\overrightarrow{AC}$ and $\overrightarrow{BC}$ as linear combinations of the vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$.
- b) Show that the angle θ between $\overrightarrow{AB}$ and $\overrightarrow{AC}$ is 90° . (Hint: Use the expressions from part a)

T1Q3. Let $\mathbf{u}, \mathbf{v}$ be vectors in $\mathbb{R}^3$ of length 1, satisfying $\mathbf{u} \cdot \mathbf{v} = 0$.

- a) Show that $\|\mathbf{u} \times \mathbf{v}\| = 1$.
- b) Show that the angle between the vector $\mathbf{u}$ and the vector $\mathbf{u} - \mathbf{u} \times \mathbf{v}$ is $\frac{\pi}{4}$.
- c) For which values of λ is it true that the vectors $\mathbf{a} := \lambda\mathbf{u} + \mathbf{v} - \mathbf{u} \times \mathbf{v}$, $\mathbf{b} := \lambda\mathbf{u} - \mathbf{v} + \mathbf{u} \times \mathbf{v}$ are orthogonal?

T1Q4. Let $\mathbf{u}, \mathbf{v}, \mathbf{w}$ be vectors in $\mathbb{R}^3$, and let $\mathbf{i} = (1, 0, 0)$ denote the unit vector in the x -direction.

- a) Is it true that if $\mathbf{u} \times \mathbf{v} = \mathbf{u} \times \mathbf{w}$ and $\mathbf{u} \neq 0$, then $\mathbf{v} = \mathbf{w}$? (Give reasons for your answer).
- b) Show that if $\mathbf{i} \cdot \mathbf{v} = \mathbf{i} \cdot \mathbf{w}$ and $\mathbf{i} \times \mathbf{v} = \mathbf{i} \times \mathbf{w}$, then $\mathbf{v} = \mathbf{w}$.

T1Q5. Let $\mathbf{u}, \mathbf{v}$ be vectors in $\mathbb{R}^3$. We denote by

$$\text{Proj}_{\mathbf{v}} \mathbf{u} := \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{v}\|^2} \mathbf{v}$$

the projection of $\mathbf{u}$ onto $\mathbf{v}$.

- a) Prove that $\|\text{Proj}_{\mathbf{v}} \mathbf{u}\| \leq \|\mathbf{u}\|$.
- b) Under what circumstances does equality hold in part a)?
- c) Prove that $(\mathbf{u} - \text{Proj}_{\mathbf{v}} \mathbf{u}) \cdot \text{Proj}_{\mathbf{v}} \mathbf{u} = 0$.

T2Q1. Consider the hyperplane H in $\mathbb{R}^4$ defined by the equation $5x_1 - 2x_2 = x_3 - 3x_4 + 5$.

- a) Write down a normal vector to H .
- b) Give a parametrisation for the hyperplane H , in the form $\mathbf{x} = \mathbf{a} + t_1\mathbf{u}_1 + t_2\mathbf{u}_2 + t_3\mathbf{u}_3$ for vectors $\mathbf{a}, \mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3$ in $\mathbb{R}^4$.

T2Q2. Find the distance between the following two lines in $\mathbb{R}^3$:

$$\begin{aligned} l_1(s) &= (-4, 9, -4) + s(4, -3, 0), \\ l_2(t) &= (5, 2, 10) + t(4, 3, 2). \end{aligned}$$

- M1Q3. a) (2 points) For vectors $\vec{v}, \vec{w} \in \mathbb{R}^n$, define the *length* $\|\vec{v}\|$ and the *angle* $\theta(\vec{v}, \vec{w})$ between $\vec{v}$ and $\vec{w}$.

Recall, we call a vector $\vec{v} \in \mathbb{R}^n$ a *unit vector* if $\|\vec{v}\| = 1$.

- b) Let $\vec{v}_1, \vec{v}_2, \vec{v}_3 \in \mathbb{R}^3$ be three unit vectors such that $\vec{v}_1 + \vec{v}_2 + \vec{v}_3 = 0$.
- i) Show that $\vec{v}_1 \cdot (\vec{v}_2 + \vec{v}_3) = -1$.
 - ii) Deduce that $\vec{v}_1 \cdot \vec{v}_2 + \vec{v}_1 \cdot \vec{v}_3 + \vec{v}_2 \cdot \vec{v}_3 = -\frac{3}{2}$.
 - iii) State the possible values of $\theta(\vec{v}_1, \vec{v}_2)$.
 - iv) Can you say anything about the triple product of $\vec{v}_1, \vec{v}_2, \vec{v}_3$? Explain your answer.

Curves, Surfaces, Coordinate Systems

- A2Q1. Consider the curve in $\mathbb{R}^2$ defined by the equation $r = 2\cos(\theta) - 4\sin(\theta)$ in polar coordinates. Find an equation in Cartesian (x, y) -coordinates for the curve, and describe the curve's shape.

- A2Q2. In this question, we denote cylindrical coordinates on $\mathbb{R}^3$ by (r, θ, z) , and spherical polar coordinates on $\mathbb{R}^3$ by (ρ, ϕ, θ) .

- a) Let $P \in \mathbb{R}^3$ be the point $(r, \theta, z) = (4\sqrt{2}, \frac{\pi}{4}, 6)$ in cylindrical coordinates. Find the Cartesian coordinates (x, y, z) of the point P .
- b) Let $Q \in \mathbb{R}^3$ be the point $(\rho, \phi, \theta) = (3, \frac{\pi}{3}, \frac{2\pi}{3})$ in spherical polar coordinates. Find the Cartesian coordinates (x, y, z) of the point Q .

- A2Q3. Let $\gamma : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be a C^1 -curve. We define the sphere of radius $r > 0$ at the origin in $\mathbb{R}^n$ by

$$\mathbb{S}_r := \partial B_r(0) = \{\vec{x} \in \mathbb{R}^n : \|\vec{x}\| = r\}.$$

Suppose that $\gamma(t)$ is orthogonal to $\gamma'(t)$ for every $t \in \mathbb{R}$. If $\gamma(0)$ is non-zero, show that the curve lies within a sphere. More precisely, show that there exists $r > 0$ such that

$$\gamma(t) \in \mathbb{S}_r, \quad \forall t \in \mathbb{R}.$$

- A2Q4. (a) Let $\vec{\gamma} : I \rightarrow \mathbb{R}^2$ be a curve. Suppose in polar coordinates we have

$$\vec{\gamma}(t) = (r(t), \theta(t)), \quad \forall t \in I.$$

Show that the speed of the curve in polar coordinates is given by the equation

$$\|\vec{\gamma}'(t)\| = \sqrt{r'(t)^2 + r^2(t)\theta'(t)^2}.$$

- (b) (3 points) For $k > 0$, the logarithmic spiral can be parameterised in polar coordinates as

$$\begin{cases} r(t) = e^{-kt} \\ \theta(t) = t, \end{cases} \quad \forall t \in [0, \infty).$$

Calculate the arc-length of this curve for each $k > 0$.

- A2Q5. (**Spherical polars in $\mathbb{R}^4$**) We may define *4D spherical polar coordinates* as follows. For a point $\vec{x} = (x, y, z, w) \in \mathbb{R}^4$, define:

- σ = distance from $\vec{x}$ to the origin,
- η = angle between $\vec{x}$ and the positive w -axis,
- ϕ = angle between $(x, y, z, 0)$ and the positive z -axis,
- θ = anticlockwise angle from the positive x -axis to $(x, y, 0, 0)$.

- a) Using the dot product formula for the angle applied to the vectors (x, y, z, w) and $(0, 0, 0, 1)$, show that $w = \sigma \cos(\eta)$.

- b) Using the formula for spherical polar coordinates applied to $(x, y, z) \in \mathbb{R}^3$ or otherwise, write down similar expressions for x, y, z in terms of $\sigma, \eta, \phi, \theta$.
- c) Express the point $\vec{x} = (2, -2, -2, 2)$ in 4D spherical polar coordinates.

T2Q3. Consider the curve $\mathbf{x} : [0, 2\pi] \rightarrow \mathbb{R}^2$, defined by

$$\mathbf{x}(t) = \left(\cos(t), \frac{1}{2} \sin(2t) \right).$$

- a) Sketch the curve. Is the curve closed? Is the curve simple?
- b) Show that for any $\mathbf{a} \in \mathbb{R}^2$, $\int_0^{2\pi} \mathbf{a} \cdot \mathbf{x}'(t) dt = 0$.
- c) More generally, prove that for any closed C^1 curve $\mathbf{y} : [a, b] \rightarrow \mathbb{R}^n$ and any vector $\mathbf{a} \in \mathbb{R}^n$,

$$\int_a^b \mathbf{a} \cdot \mathbf{y}'(t) dt = 0.$$

T2Q4. Let $\mathbf{x}(t) : [t_0, t_1] \rightarrow \mathbb{R}^3$ be a C^1 curve. We define $\mathbf{y}(s) : [s_0, s_1] \rightarrow \mathbb{R}^3$ to be the re-parameterization of $\mathbf{x}(t)$ through the strictly increasing C^1 bijection $s : [t_0, t_1] \rightarrow [s_0, s_1]$. In other words, $\mathbf{y}(s(t)) = \mathbf{x}(t)$. Prove that $\mathbf{x}(t)$ and $\mathbf{y}(s)$ are of the same length.

T2Q5. Consider the curve C_1 in $\mathbb{R}^2$ defined by the following equation in (x, y) -coordinates:

$$25x^2 + 144y^2 = 169.$$

- a) Write down an equation for the curve using the polar coordinate functions r and θ , of the form $r = f(\theta)$.
- b) Use this equation to deduce the minimum and maximum values of the radial function r on the curve.

Now consider the curve C_2 in $\mathbb{R}^2$ with an (x, y) -coordinate parametrization as follows:

$$x = \cos(t), \quad y = \frac{1}{2} \sin(2t).$$

- c) Show that the polar coordinate equation $r^2 = \frac{\cos(2\theta)}{\cos(\theta)^4}$ defines the same curve.
- d) Find the subset of values of θ between 0 and 2π for which there exists a point on the curve C_2 with angle θ .

T2Q6. Consider the two surfaces P_1, P_2, P_3 in $\mathbb{R}^3$ defined by the following equations in cylindrical polar coordinates:

$$\begin{aligned} S_1 : \quad r^2 &= z(1-z), \\ S_2 : \quad r &= \lambda z. \end{aligned}$$

- a) Show that the intersection of S_1 and S_2 away from the origin is a circle, and determine its radius.
- b) By expressing the cylindrical coordinate functions (r, θ, z) in terms of the spherical coordinate functions (ρ, θ, ϕ) , express the surfaces S_1 and S_2 using equations in the spherical coordinates (ρ, θ, ϕ) .

P1Q2. a) State the Cauchy-Schwarz inequality.

- b) For any four real numbers $\alpha_1, \alpha_2, \alpha_3, \alpha_4 \in \mathbb{R}$, show that the following inequality holds

$$\frac{\alpha_1 + \alpha_2 + \alpha_3 + \alpha_4}{2} \leq (\alpha_1^2 + \alpha_2^2 + \alpha_3^2 + \alpha_4^2)^{\frac{1}{2}}.$$

c) Consider the curve $\gamma : [0, 1] \rightarrow \mathbb{R}^4$ given by

$$\gamma(t) = \begin{pmatrix} t + \sin^2(t) + \cos^3(t) - 2\sin(t)\cos^2(t) \\ -3t + \cos^2(t) + \cos^5(t)\sin^3(t) + 2\sin(t)\cos^2(t) \\ 2t - \cos^3(t) - \sin^2(t) \\ 2t - \cos^5(t)\sin^3(t) - \cos^2(t) \end{pmatrix}, \quad \forall t \in [0, 1].$$

If S denotes the arclength of γ , using your result from part b) or otherwise, show that $S \geq 1$.

P1Q3. In this question, we denote cylindrical coordinates on $\mathbb{R}^3$ by (r, θ, z) .

- Let $P \in \mathbb{R}^3$ be the point $(r, \theta, z) = (4\sqrt{2}, \frac{\pi}{4}, 6)$ in cylindrical coordinates. Find the Cartesian coordinates (x, y, z) of the point P .
- For each pair of integers $\alpha, \beta \in \mathbb{Z}$, consider the curve $\gamma_{\alpha, \beta} : [0, 1] \rightarrow \mathbb{R}^3$, given in cylindrical coordinates by the parameterisation

$$\begin{cases} r(t) = 10 + \sin(\frac{\pi}{2}(\alpha + \beta)t), \\ \theta(t) = \pi\beta t, \\ z(t) = 5\cos(2\pi\alpha t), \end{cases} \quad \forall t \in [0, 1].$$

For which pairs of integers $\alpha, \beta \in \mathbb{Z}$ is the curve $\gamma_{\alpha, \beta}$ closed?

- Is the curve $\gamma_{2024, 2}$ simple? Justify your answer.
- Is the curve $\gamma_{4, 4}$ simple? Justify your answer.

P1Q7. Consider a curve $\gamma : I \rightarrow \mathbb{R}^3$ parameterised in spherical coordinates

$$\gamma(t) = (\rho(t), \phi(t), \theta(t)), \quad t \in I.$$

- Using the formulas for the Cartesian coordinates

$$\begin{aligned} x &= \rho \sin \phi \cos \theta, \\ y &= \rho \sin \phi \sin \theta, \\ z &= \rho \cos \phi, \end{aligned}$$

find equations for $x'(t)$, $y'(t)$, and $z'(t)$ in terms of $\rho'(t)$, $\phi'(t)$, and $\theta'(t)$.

- Substituting your answers from a) into the formula

$$\|\gamma'(t)\| = \sqrt{x'(t)^2 + y'(t)^2 + z'(t)^2},$$

find an equation for the speed of the curve in spherical coordinates.

- Using the formula from b) or otherwise, calculate the arclength of the curve

$$\begin{cases} \rho(t) = e^t, \\ \theta(t) = \log(\tan t), \\ \phi(t) = 2t, \end{cases} \quad \forall t \in \left(\frac{\pi}{4}, \frac{\pi}{2}\right).$$

P2Q2. Consider the two surfaces in $\mathbb{R}^3$ given by the equations

$$S_1 = \{2x^2 - 2y^2 + z^2 = 0\}, \quad S_2 = \{x^2 + y^2 - 2z^2 = 8\}.$$

- Find an equation of the tangent plane to S_1 at the point $(1, 1, 0)$.
- Find the point(s) on S_2 whose tangent plane(s) are parallel to the plane you calculated in part a).

Topology

T3Q1. a) Let $\vec{x}_0 \in \mathbb{R}^n$ and $r > 0$. Show that the set

$$\overline{B_r(\vec{x}_0)} = \{\vec{x} \in \mathbb{R}^n : \|\vec{x} - \vec{x}_0\| \leq r\},$$

is a closed set.

This justifies naming $\overline{B_r(\vec{x}_0)}$ a closed ball.

b) Show that, for a pair of open sets X_1, X_2 , their *union*

$$X_1 \cup X_2 := \{\vec{x} \in \mathbb{R}^n : \vec{x} \in X_1 \text{ or } \vec{x} \in X_2\}$$

is also an open set.

c) Show that, for a pair of open sets X_1, X_2 , their *intersection*

$$X_1 \cap X_2 := \{\vec{x} \in \mathbb{R}^n : \vec{x} \in X_1 \text{ and } \vec{x} \in X_2\}$$

is also an open set.

d) Suppose now we have a (not necessarily finite) collection of open sets

$$\{X_\alpha : \alpha \in \mathcal{A}\}.$$

Is their union $U := \bigcup_{\alpha \in \mathcal{A}} X_\alpha$ necessarily open? Justify your answer.

e) Is their intersection $C := \bigcap_{\alpha \in \mathcal{A}} X_\alpha$ necessarily open? Justify your answer.

T3Q2. Let $A \subseteq \mathbb{R}^n$ and consider the set of vectors whose distance to A is zero:

$$\{\vec{x} \in \mathbb{R}^n : \forall \epsilon > 0, \exists \vec{a} \in A \text{ such that } \|\vec{x} - \vec{a}\| < \epsilon\}.$$

Show that this set is the same as the closure of A :

$$\overline{A} = \text{Int}(A) \cup \partial A.$$

M1Q4. For each of the following sets, state whether the set is:

(i) open, (ii) closed, (iii) bounded, (iv) path connected.

You do not need to justify your reasoning.

a) $S_1 = \{x \in \mathbb{R}^{100} : \|x\| < 1\}.$

b) $S_2 = \{(x, y) \in \mathbb{R}^2 : |y| \geq 1 + x^2\}.$

c) $S_3 = \mathbb{Q} = \{\frac{p}{q} \in \mathbb{R} : p, q \in \mathbb{Z}, q \neq 0\}.$

d) $S_4 = \{(x, y) \in \mathbb{R}^2 : 0 < y \leq \frac{1}{1+x^2}\} \cup \{(0, 0)\}.$

P1Q4. For each of the following sets, state whether the set is:

(i) open, (ii) closed, (iii) bounded, (iv) path connected.

You do not need to justify your reasoning.

a) $S_1 = \mathbb{R}^n$ (as a subset of itself).

b) $S_2 = \{(x, y, z) \in \mathbb{R}^3 : x + 4y - 2z = 6\}.$

c) $S_3 = \{(x, y) \in \mathbb{R}^2 : |x| + |y| \leq 10\}.$

d) $S_4 = \{(x, y, z) \in \mathbb{R}^3 : 1 < e^{x^2+y^2} < 2\}.$

Level Sets and Graphs

M1Q2. Consider the function $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ defined by $f(x, y, z) := \sqrt{x^2 + y^2 + z^2}$.

- Explain why the point $(1, 2, 3, 4)$ is not an element of $\text{Graph}(f)$.
- For which value of λ is $(1, 1, 1, \lambda)$ an element of $\text{Graph}(f)$?
- Write down an equation describing the level set L_c of f at the value $c \in \mathbb{R}$.
- For which value of c is the vector $(1, 1, 1)$ an element of L_c ?

P1Q1. For this question, suppose $v_1, v_2, v_3 \in \mathbb{R}^3 \setminus \{0\}$ are three non-zero vectors which sum to zero

$$v_1 + v_2 + v_3 = 0.$$

Assume also that these three vectors are pairwise non-parallel, i.e v_i is not parallel to v_j for any $i, j \in \{1, 2, 3\}$.

- State the triangle inequality for vectors in $\mathbb{R}^n$, making sure to mention when equality holds.
- Show that

$$\|v_1\| < \|v_2\| + \|v_3\|.$$

- Consider the function $f : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ defined by

$$f(x, y, z) = \begin{pmatrix} f_1(x, y, z) \\ f_2(x, y, z) \\ f_3(x, y, z) \end{pmatrix} = \begin{pmatrix} v_1 \cdot (x, y, z) \\ v_2 \cdot (x, y, z) \\ v_3 \cdot (x, y, z) \end{pmatrix}.$$

Describe geometrically the level sets of the first component function $f_1 : \mathbb{R}^3 \rightarrow \mathbb{R}$.

- For which vectors $c = \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix} \in \mathbb{R}^3$ is the level set of f at c non-empty?

When the level set is non-empty, give a geometric description of the level set.

Limits, Continuity

A3Q1. Evaluate the following limits, or show they do not exist, with full justification:

- $\lim_{z \rightarrow 3} \left(\lim_{y \rightarrow 2} \left(\lim_{x \rightarrow 1} \frac{e^{x+y} - 1}{\sin(2x-y) + z^2} \right) \right)$
- $\lim_{(x,y) \rightarrow (0,0)} \frac{\sin(xy^4)}{x^2 + y^2}$
- $\lim_{(x,y) \rightarrow (1,1)} \frac{xy - x - y + 1}{|xy - x - y + 1|}$
- $\lim_{(x,y) \rightarrow (0,0)} f(x, y)$ where $f : \mathbb{R}^2 \setminus \{(x, y) : x = y\} \rightarrow \mathbb{R}$, $f(x, y) = \frac{x^2y + y^2x + x^3}{x^2 - 2xy + y^2}$

A3Q2. Using the (ε, δ) -definition of continuity directly, show that the function $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ given by $f(x, y, z) = \frac{x+y+z}{x^2+y^2+z^2+1}$ is continuous at the origin $(0, 0, 0)$.

A3Q3. Show that the function $F : \mathbb{R}^3 \rightarrow \mathbb{R}$ defined by

$$F(x, y, z) = \begin{cases} \frac{\exp(\frac{1}{yz-x^3})}{(x^3-yz)} & : x^3 - yz > 0 \\ 0 & : x^3 - yz \leq 0 \end{cases},$$

is continuous everywhere in $\mathbb{R}^3$. (*Hint: Use the fact that the composition of continuous functions is continuous*)

A3Q4. Consider the rational function

$$Q(x, y) = \frac{x^6 - x^5y + xy^5 - y^6}{xy^3 - x^3y}$$

What is the largest subset of the plane $\Omega \subseteq \mathbb{R}^2$ on which we can extend Q to a continuous function? Justify your answer.

A3Q5. We say a real number is rational if it can be expressed as a ratio of two integers. We denote the collection of all rational numbers as

$$\mathbb{Q} = \left\{ \frac{p}{q} : p, q \in \mathbb{Z}, q \neq 0 \right\}.$$

At which points in $\mathbb{R}^3$ is the following function continuous?

$$f(x, y, z) := \begin{cases} x + y - z & : x, y, z \in \mathbb{Q} \\ 1 & : \text{otherwise} \end{cases}.$$

Justify your answer.

T3Q3. For the following examples, does the limit exist, and if so, what is its value?

a) $\lim_{(x,y) \rightarrow (0,0)} \frac{\sin(xy)}{xy}$

b) $\lim_{(x,y) \rightarrow (0,0)} \frac{\sin(xy)}{x+y}$

c) $\lim_{(x,y) \rightarrow (0,0)} \frac{\sin(xy)}{x^2+y^2}$

T3Q4. In this question, consider the function $f : \mathbb{R}^3 \rightarrow \mathbb{R}^2$, defined by

$$f(x, y, z) = \begin{pmatrix} f_1(x, y, z) \\ f_2(x, y, z) \end{pmatrix} = \begin{pmatrix} (x-4)^2 + (y-3)^2 + z^2 \\ x - y + 2z \end{pmatrix}.$$

a) Describe the level sets of the first component function

$$f_1(x, y, z) = (x-4)^2 + (y-3)^2 + z^2.$$

b) Describe the level sets of the second component function

$$f_2(x, y, z) = x - y + 2z.$$

c) Describe the level set of f at the point $(1, 1) \in \mathbb{R}^2$.

T3Q5. Suppose $\Omega \subseteq \mathbb{R}^n$ is open, $\vec{f}, \vec{g} : \Omega \rightarrow \mathbb{R}^n$, and $\vec{a} \in \Omega$. If $\lim_{\vec{x} \rightarrow \vec{a}} \vec{f}(\vec{x}) = 0$ and $\vec{g}(\vec{x})$ is bounded, deduce that $\lim_{\vec{x} \rightarrow \vec{a}} \vec{f}(\vec{x}) \cdot \vec{g}(\vec{x}) = 0$.

T4Q1/M1Q1. Consider the function $\mathbf{f} : \mathbb{R}^2 \setminus \{(x, y) : x = 1 \text{ or } y = 1\} \rightarrow \mathbb{R}^2$, defined by

$$\mathbf{f}(x, y) := \left(\frac{(y^2-1)(x^2-1)}{(y-1)(x-1)}, e^{x+y} \right).$$

a) Calculate the following limits, with full justification:

i) $\lim_{(x,y) \rightarrow (0,0)} \mathbf{f}(x, y),$

ii) $\lim_{(x,y) \rightarrow (1,1)} \mathbf{f}(x, y).$

b) Can $\mathbf{f}$ be extended to a continuous function $\bar{\mathbf{f}} : \mathbb{R}^2 \rightarrow \mathbb{R}^2$? Explain why or why not.

M1Q5. Consider the two functions defined by:

$$f : \Omega_1 := \mathbb{R}^2 \setminus \{y = 0\} \rightarrow \mathbb{R}, \quad g : \Omega_2 := \mathbb{R}^2 \setminus \{y = 1\} \rightarrow \mathbb{R}$$

$$f(x, y) := \frac{x^2 + y}{y}, \quad g(x, y) := \frac{x^2 y^2 - 1}{y - 1}.$$

- a) By considering the value of $g(x, y)$ along several different straight lines through $(1, 1)$, show that the limit

$$\lim_{(x,y) \rightarrow (1,1)} g(x, y)$$

does not exist.

- b) By considering the value of $f(x, y)$ along suitable lines, show that the limit

$$\lim_{(x,y) \rightarrow (0,0)} f(x, y)$$

does not exist.

M1Q6. In this question, we consider the function

$$f : \Omega \rightarrow \mathbb{R}, \quad \Omega = \{(x, y) : y \neq 0, -2\pi y < x < 2\pi y\}, \quad f(x, y) := \frac{x - 1}{\cos\left(\frac{x}{4y}\right)}.$$

- a) Show that if $\|(x, y) - (1, 1)\| < \frac{1}{2}$, then $0 < \frac{x}{4y} < \frac{\pi}{4}$.
 b) Deduce using the sandwich/squeeze theorem or otherwise that

$$\lim_{(x,y) \rightarrow (1,1)} f(x, y) = 0.$$

P1Q5. Show that the function $F : \mathbb{R}^3 \rightarrow \mathbb{R}$ defined by

$$F(x, y, z) = \begin{cases} \frac{\exp(\frac{1}{yz-x^3})}{(x^3-yz)} & : x^3 - yz > 0 \\ 0 & : x^3 - yz \leq 0 \end{cases},$$

is continuous everywhere in $\mathbb{R}^3$.

Differentiability, Derivatives for Scalar-Valued Functions

A4Q1. Consider the function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ given by

$$f(x, y) = \begin{cases} \frac{x^3+y^3}{x^2+y^2} & \text{if } (x, y) \neq (0, 0), \\ 0 & \text{if } (x, y) = (0, 0). \end{cases}$$

- a) Show that f is continuous everywhere in $\mathbb{R}^2$.
 b) Calculate $\frac{\partial f}{\partial x}$ at the point $(0, 0)$ using the limit definition of the partial derivative.
 c) Calculate $\frac{\partial f}{\partial x}$ at a point $(x, y) \neq (0, 0)$ by treating f as a single-variable function in x , keeping y fixed.
 d) Conclude that $\frac{\partial f}{\partial x}$ exists everywhere, but is not continuous everywhere.

A4Q2. Consider the function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$, defined by

$$f(x, y) := \tan^{-1}(e^{x+y}).$$

- a) Without calculating any derivatives, explain how we know that f is differentiable everywhere.
 b) Find the rate of change of f at $(0, \ln(2))$ in the direction of $\vec{v} = (1, 2)$.

A4Q3. Consider the function $T : \mathbb{R}^3 \rightarrow \mathbb{R}$ describing the temperature (in degrees Celsius $^{\circ}C$) in a room (measured in metres m), defined by

$$T(x, y, z) := xy + yz + xz.$$

A mosquito is currently at the point $(1, 1, 1)$.

- Calculate $\frac{\partial T}{\partial x}, \frac{\partial T}{\partial y}, \frac{\partial T}{\partial z}$, and write down the gradient $\vec{\nabla}T$ of T at the point $(1, 1, 1)$.
- In which direction $\vec{u}$ (i.e. such that $\|\vec{u}\| = 1$) should the mosquito move to increase its temperature most quickly?
- Is there any direction $\vec{u}$ the mosquito can move in such that the rate of temperature increase it experiences is $5^\circ\text{C}/m$? (No explanation is necessary.)
- Write down the linear approximation of T at $(1, 1, 1)$. Use this to estimate the change in temperature the mosquito experiences as it moves to the point $(1.01, 1.01, 0.99)$.

A4Q4. Consider the function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$, defined by

$$f(x, y) := xe^{2x+2y}.$$

- Calculate the partial derivatives $\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}$.
- Show by mathematical induction or otherwise that for $n, m \geq 0$,

$$\begin{aligned}\frac{\partial^n f}{\partial x^n} &= 2^n x e^{2x+2y} + n 2^{n-1} e^{2x+2y}, \\ \frac{\partial^{n+m} f}{\partial x^n \partial y^m} &= 2^{m+n} x e^{2x+2y} + n 2^{n+m-1} e^{2x+2y}.\end{aligned}$$

- Explain why f is a C^∞ function.

A4Q5. Consider the function $f : \mathbb{R}^3 \rightarrow \mathbb{R}$, defined by

$$f(x, y, z) = \begin{cases} \frac{xyz}{x^2+y^2+z^2} & \text{if } (x, y, z) \neq (0, 0, 0) \\ 0 & \text{if } (x, y, z) = (0, 0, 0) \end{cases}.$$

- Calculate $D_{\vec{u}}f(0, 0, 0)$ for $\vec{u} = (\sin(\theta)\cos(\phi), \sin(\theta)\sin(\phi), \cos(\theta))$.
- Calculate $\vec{\nabla}f(0, 0, 0)$. (*Hint: You could use the fact that $\frac{\partial f}{\partial x_i} = D_{\vec{e}_i}f$*)
- Using facts relating the gradient and directional derivative, or otherwise, show that f is not differentiable at $(0, 0, 0)$.

A4Q6. Let $F(x, y, z) = x^3 + y^3 + z^3 - 3xyz - 4$. It is given that $F(x, y, z) = 0$ defines an implicit function $z = f(x, y)$ near the point $(1, 1, 2)$. Find $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$ at $(1, 1, 2)$.

T4Q2. For the following functions, find $\frac{\partial f}{\partial x}(0, 0)$ and $\frac{\partial f}{\partial y}(0, 0)$, or show they do not exist.

- $f_1 : \mathbb{R}^2 \rightarrow \mathbb{R}, f_1(x, y) = \cos^2(3x - y^2)$,
- $f_2 : \mathbb{R}^2 \rightarrow \mathbb{R}, f_2(x, y) = 3x^y$,
- $f_3 : \mathbb{R}^2 \rightarrow \mathbb{R}$,

$$f_3(x, y) := \begin{cases} \frac{\sin(x^3+y^3)}{x^2+y^2} & \text{for } (x, y) \neq (0, 0), \\ 0 & \text{for } (x, y) = (0, 0). \end{cases}$$

T4Q3. Consider the function

$$f : \mathbb{R}^2 \rightarrow \mathbb{R}, \quad f(x, y) := x^2 \sin(x) \cos(y).$$

Without calculating any derivatives, explain how you know that $f_{xxxyyy} = f_{xyxyxy}$.

T4Q4. Consider the function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ defined by

$$f(x, y) = \begin{cases} \frac{x^3}{x^2+y^2} & : (x, y) \neq (0, 0) \\ 0 & : (x, y) = (0, 0) \end{cases}.$$

- Calculate the directional derivative $D_u f(0, 0)$ for the vector $u = (\cos \theta, \sin \theta)$.
- Calculate the gradient $\nabla f(0, 0)$.
- Is the function f differentiable at the origin? Justify your answer.

T4Q5. Consider the family of functions

$$f_\alpha(x, y, z) = e^x + y \cos z + \alpha(x^3 - y + e^{-z} \sin x), \quad \forall \alpha \in \mathbb{R}.$$

For which values of α does the function increase most rapidly in a direction parallel to the x -axis at the origin? Justify your answer.

T4Q6. In this question, we show that if a multivariable function f has a good linear approximation, then it satisfies the definition of differentiability.

Let $\Omega \subseteq \mathbb{R}^n$ open, $f : \Omega \rightarrow \mathbb{R}$ and $\vec{a} \in \Omega$. We assume that there exists some affine function $L : \mathbb{R}^n \rightarrow \mathbb{R}$, defined by

$$L(\vec{x}) = \lambda + \vec{\alpha} \cdot (\vec{x} - \vec{a}), \quad (\lambda \in \mathbb{R}, \alpha \in \mathbb{R}^n)$$

such that the error function $\varepsilon(\vec{x}) = f(\vec{x}) - L(\vec{x})$ satisfies

$$\lim_{\vec{x} \rightarrow \vec{a}} \frac{\varepsilon(\vec{x})}{\|\vec{x} - \vec{a}\|} = 0.$$

- Show that $\lim_{\vec{x} \rightarrow \vec{a}} f(\vec{x})$ exists and equals λ .
- Show that $\frac{\partial L}{\partial x_i}(\vec{a})$ exists and equals α_i .
- Show that all the partial derivatives $\frac{\partial f}{\partial x_i}(a)$ (for $1 \leq i \leq n$) exist, and find their value.

M1Q7. Consider the following *one-dimensional heat equation*, for a function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$, where the coordinates on $\mathbb{R}^2$ are (x, t) :

$$\frac{\partial f}{\partial t} = \frac{\partial^2 f}{\partial x^2}.$$

For constants $\alpha, \beta \in \mathbb{R}$, define the function

$$u(x, t) := \sin(\alpha x) e^{-\beta t}.$$

Find a relationship between the constants α and β for this function to be a solution to the one-dimensional heat equation.

M1Q8. Consider the function $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ given by

$$f(x, y, z) := \begin{cases} x \sin\left(\frac{y}{z}\right) & \text{if } z \neq 0, \\ 0 & \text{otherwise.} \end{cases}$$

Is f differentiable at the point $(0, 0, 0)$? Explain why or why not, with full justification.

P1Q6. Suppose $\Omega \subseteq \mathbb{R}^n$ is open, $f, g : \Omega \rightarrow \mathbb{R}$, and $a \in \Omega$. Define a new function $h : \Omega \rightarrow \mathbb{R}$ given by the product of f and g ,

$$h(x) := f(x)g(x), \quad \forall x \in \Omega.$$

- Define what it means for the function f to be differentiable at a
- Suppose f is differentiable at a with $f(a) = 0$ and $\nabla f(a) = 0$. If g is continuous at a , show that the function $h(x)$ is differentiable at a .
- If we remove the hypothesis that $f(a) = 0$, is the result from part c) still true? That is, if f is differentiable at a with $\nabla f(a) = 0$, and g is continuous at a , is the function $h(x)$ necessarily differentiable at a ? Either provide a proof or a counterexample.

Jacobian Matrix, Chain Rule

A4Q7. Consider the following functions, where the domain is unspecified:

$$\vec{F} : \Omega_1 \subset \mathbb{R}^3 \rightarrow \mathbb{R}^2, \quad \vec{F}(x, y, z) := \begin{pmatrix} e^x \sin(y) + \sec(x) \\ \frac{x^2 y}{z-1} + \cosh(z) \end{pmatrix},$$

$$\vec{G} : \Omega_2 \subset \mathbb{R}^2 \rightarrow \mathbb{R}^2, \quad \vec{G}(u, v) = \begin{pmatrix} u^2 v^2 - 2uv \\ \ln(\frac{u}{2v}) \end{pmatrix}.$$

Recall that we denote the i -th component of a vector valued function f by f_i , so for example $F_1(x, y, z) = e^x \sin(y) + \sec(x)$.

a) Let $\vec{a} := \begin{pmatrix} 0 \\ \frac{\pi}{2} \\ 0 \end{pmatrix}$. Explain why the composition $\vec{G} \circ \vec{F}$ is defined near $\vec{a}$.

b) By using the chain rule or otherwise, calculate the Jacobian matrix of $\vec{G} \circ \vec{F}$ at $\vec{a}$.

c) What is the value of $\frac{\partial(\vec{G} \circ \vec{F})_1}{\partial x}(\vec{a})$?

T5Q1. Let $w(x, y, z) = x^2 e^{2y} \cos 3z$. Find the value of $\frac{dw}{dt}$ at the point $(1, \ln 2, 0)$ on the curve $\gamma(t) = (\cos t, \ln(t+2), t)$.

T5Q2. Suppose that we substitute polar coordinates $x = r \cos \theta$ and $y = r \sin \theta$ in a differentiable function $w = f(x, y)$.

a) Show that

$$\frac{\partial w}{\partial r} = f_x \cos \theta + f_y \sin \theta$$

and

$$\frac{1}{r} \frac{\partial w}{\partial \theta} = -f_x \sin \theta + f_y \cos \theta.$$

b) Solve the equations in part (a) to express f_x and f_y in terms of $\partial w / \partial r$ and $\partial w / \partial \theta$.

c) Show that

$$(f_x)^2 + (f_y)^2 = \left(\frac{\partial w}{\partial r} \right)^2 + \frac{1}{r^2} \left(\frac{\partial w}{\partial \theta} \right)^2$$

T5Q3. Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be a C^2 -function. We define the Laplace operator to be

$$\Delta f = \sum_{i=1}^n \frac{\partial^2 f}{\partial x_i^2}.$$

Assume that f is a radial function, i.e., $f(x) = \rho(r)$, $r = |x|$. Show that

$$\Delta f(x) = \rho''(r) + \frac{n-1}{r} \rho'(r).$$

Hint: From the chain rule, $\frac{\partial \rho(r)}{\partial x_i} = \rho'(r) \cdot \frac{\partial r}{\partial x_i}$.

Finding and Classifying Extrema, Taylor Polynomials

T5Q4. a) Suppose $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is a continuous function such that $\lim_{x \rightarrow \infty} f(x) = \infty$. That is

$$\forall C \in \mathbb{R}, \exists R > 0 \quad \text{such that} \quad \|x\| \geq R \implies f(x) > C.$$

Show that f attains a global minimum on $\mathbb{R}^n$.

- b) Suppose $g : \mathbb{R}^n \rightarrow (0, \infty)$ is a positive continuous function such that $\lim_{x \rightarrow \infty} g(x) = 0$. That is

$$\forall \epsilon > 0, \exists R > 0 \text{ such that } \|x\| \geq R \implies g(x) < \epsilon.$$

Show that g attains a global maximum on $\mathbb{R}^n$.

- c) Does the function $g : \mathbb{R}^n \rightarrow (0, \infty)$ from part b) necessarily attain a global minimum? Justify your answer.
- d) Find the global maximum of the function

$$h(x, y) = \frac{1 + |x| + |y|}{1 + x^2 + y^2}, \quad \forall (x, y) \in \mathbb{R}^2.$$

T5Q5. For each of the following functions with the given domain:

- State whether the extreme value theorem (EVT) applies, with full explanation,
- Find all global maxima, or show there are none,
- Find all global minima, or show there is none.

a) $f_1 : [-1, 1] \rightarrow \mathbb{R}, \quad f_1(x) = \begin{cases} x + 1 & x \leq 0, \\ 1 - x & x > 0. \end{cases}$

b) $f_2 : [-1, 1] \rightarrow \mathbb{R}, \quad f_2(x) = \begin{cases} x + 1 & x \leq 0, \\ -x & x > 0. \end{cases}$

c) $f_3 : B_\pi(0) \rightarrow \mathbb{R}, \quad f_3(x, y) := 1 - \sin(\frac{1}{2}\sqrt{x^2 + y^2}).$

d) $f_4 : \mathbb{R}^2 \setminus \{(0, 0)\} \rightarrow \mathbb{R}, \quad f_4(x, y) := \frac{\tan^{-1}(x) - \tan^{-1}(y)}{x^2 + y^2}.$

e) $f_5 : [-1, 1] \times [-1, 1] \times [-1, 1] \subset \mathbb{R}^3 \rightarrow \mathbb{R}, \quad f_5(x, y, z) := |x| - 3|y| + 2|z|.$

T5Q6. Consider the closed domain given by

$$\Omega := \{(x, y) : x \geq 0, y \geq 0, x + y \leq 1\},$$

and the function $f : \Omega \rightarrow \mathbb{R}$ given by

$$f(x, y) := \frac{x + y - 1}{(x + 1)(y + 1)}.$$

- Explain why the function attains a global maximum and global minimum on Ω .
- Find all global maxima and global minima of f on Ω , with full justification.
- Do there exist any local maxima/minima on the interior of Ω which are not global maxima/minima? Explain why/why not.

T5Q7. In this question, we consider the function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$, defined by

$$f(x, y) := \sin(x) \sin(y) e^{-(x^2 + y^2)}.$$

Throughout, you may use the following facts freely:

- The *intermediate value theorem*: If $f : [a, b] \rightarrow \mathbb{R}$ is continuous and $f(a) < s < f(b)$, then there exists $c \in (a, b)$ such that $f(c) = s$.
- Rolle's theorem*: If $f : [a, b] \rightarrow \mathbb{R}$ is differentiable and $f(a) = f(b)$, then there exists $c \in (a, b)$ such that $f'(c) = 0$.
- Bounds on the famous constants e and π : $2 < e < 3, 3 < \pi < \sqrt{10}$.

- a) Show that there is a *unique* real number $a \in (0, \pi)$ for which $\tan(a) = \frac{1}{2a}$. Furthermore, show that:

i) $a < \frac{\pi}{4},$

- ii) $\sin(a) = \frac{1}{\sqrt{1+4a^2}}$.
- b) Show that outside the closed ball $\overline{B}_\pi(0)$, we have $f(x, y) < \frac{1}{500}$.
- c) Find all local extrema of $f(x, y)$ in $\overline{B}_\pi(0)$, and state the value of f at each one (in terms of the number a from part a) where necessary)
- d) Show by estimating that $f(a, a) > \frac{1}{500}$.
- e) Using the previous parts of the question, show that the only global maxima of f on $\mathbb{R}^2$ are at (a, a) and $(-a, -a)$.

P2Q3. For any three positive real numbers $a, b, c > 0$, consider the ellipsoid

$$E(a, b, c) = \{(x, y, z) \in \mathbb{R}^3 : x^2/a^2 + y^2/b^2 + z^2/c^2 = 1\}.$$

The temperature is given by the function $T : \mathbb{R}^3 \rightarrow \mathbb{R}$,

$$T(x, y, z) = x + 2y + z^2, \quad \forall (x, y, z) \in \mathbb{R}^3.$$

- a) Prove that T attains a maximum value on the ellipsoid $E(4, 2, 2)$, and find this maximum value of T .
- b) Let $\gamma : (-1, 1) \rightarrow \mathbb{R}^3$ be the trajectory of a particle given by the parameterisation

$$\begin{cases} x(t) = t, \\ y(t) = 2\sqrt{1-t^2} \cos\left(\frac{2\pi}{1-t^2}\right), \\ z(t) = 2\sqrt{1-t^2} \sin\left(\frac{2\pi}{1-t^2}\right), \end{cases} \quad \forall t \in (-1, 1),$$

in Cartesian coordinates. Show that the particle always lies on some ellipsoid. That is, there exists $a, b, c > 0$ such that,

$$\gamma(t) \in E(a, b, c), \quad \text{for any time } t \in (-1, 1).$$

- c) At time $t = 0$, is the temperature of the particle increasing or decreasing? Justify your answer.
- d) Does the particle achieve a global maximum temperature for times $t \in (-1, 1)$? Justify your answer.

(Note: if a maximum does exist, you do not need to find it.)

P2Q4. For this question, let $f : \Omega \subseteq \mathbb{R}^2 \rightarrow \mathbb{R}$ be the function

$$f(x, y) = x^y + y^x - x - y, \quad \forall (x, y) \in \Omega,$$

for the open subset of the plane $\Omega = \{(x, y) \in \mathbb{R}^2 : x, y > 0\}$.

- a) Show that the function f is a C^1 function and has a critical point at $(1, 1)$.
- Hint: $a^b = \exp(b \log a)$.*
- b) Show that f is a C^2 function and calculate the 2^{nd} -order Taylor polynomial of f at $(1, 1)$.
- c) Classify the type of critical point f has at $(1, 1)$.

P2Q7. Consider the function $F : \mathbb{R}^2 \rightarrow \mathbb{R}$ defined by

$$F(x, y) = \sin(x) \sin(y), \quad \forall (x, y) \in \mathbb{R}^2.$$

- a) Find and classify the critical points of F .
- b) At each critical point, find the 2^{nd} -order Taylor polynomial P_2 .

P2Q9. Let $f(x, y) = 2x^3 + 6xy^2 - 3x^2 + 3y^2$.

- a) Let $\Omega = \{x^2 + y^2 \leq 1\}$. Find all critical points of $f(x, y)$ in $\text{Int}(\Omega)$, and determine the nature of each of the critical points (i.e., local maximum, local minimum, saddle).
- b) Use the method of Lagrange multipliers to find the extrema of $f(x, y)$ restricted to $\partial\Omega$.
- c) Find all global maxima and minima of $f(x, y)$ in Ω .

Implicit Function Theorem, Inverse Function Theorem

P2Q5. Consider the function $\mathbf{F} : \mathbb{R}^2 \rightarrow \mathbb{R}^2$, defined by

$$\mathbf{F}(x, y) = (xy(x + y), xy(x - y)), \quad \forall x, y \in \mathbb{R}.$$

- a) For which values of $(x, y) \in \mathbb{R}^2$ does $\mathbf{F}$ admit a C^1 local inverse near (x, y) ?
- b) Consider now the function $\mathbf{G} : \mathbb{R}^4 \rightarrow \mathbb{R}^2$, defined by

$$\mathbf{G}(x, y) = (x \cdot y)(x + y), \quad \forall x, y \in \mathbb{R}^2.$$

That is, writing $x = (x_1, x_2), y = (y_1, y_2) \in \mathbb{R}^2$, we have

$$\mathbf{G}(x_1, x_2, y_1, y_2) = \begin{pmatrix} (x_1 y_1 + x_2 y_2)(x_1 + y_1) \\ (x_1 y_1 + x_2 y_2)(x_2 + y_2) \end{pmatrix}.$$

Show that

$$\frac{\partial \mathbf{G}}{\partial y}(x, y) = (x \cdot y)I_2 + \begin{pmatrix} x_1(x_1 + y_1) & x_2(x_1 + y_1) \\ x_1(x_2 + y_2) & x_2(x_2 + y_2) \end{pmatrix},$$

where $I_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ is the 2×2 identity matrix.

Now, let $a, b, c \in \mathbb{R}^2$ with $\mathbf{G}(a, b) = c$.

- c) If $a \cdot b > 0$, show that for solutions to the constraint $\mathbf{G}(x, y) = c$, we can locally express y as a C^1 function of x near (a, b) .
- d) If $a \cdot b < 0$ with $a > 2b$, show that for solutions to the constraint $\mathbf{G}(x, y) = c$, we can locally express y as a C^1 function of x near (a, b) .
- e) Show that, in the case $a = b = c = (0, 0) \in \mathbb{R}^2$, for solutions to the constraint $\mathbf{G}(x, y) = c$, there is no unique way to express y as a function of x locally near (a, b) .

P2Q6. Let $S \subset \mathbb{R}^3$ be defined by the following equations:

$$\begin{aligned} xy + ze^x \sin(y + z) &= 0, \\ x^2 + y^2 + z^2 &= 2. \end{aligned}$$

- a) Determine whether on S , x, y can be expressed as differentiable functions in z locally near $(0, 1, -1)$. Justify your answer.
- b) Determine whether on S , y, z can be expressed as differentiable functions in x locally near $(0, 1, -1)$. Justify your answer.

P2Q8. Let $\mathbf{F} : \mathbb{R}^4 \rightarrow \mathbb{R}^2$ be C^1 and

$$\mathbf{F}(x, y, z, w) = (F_1(x, y, z, w), F_2(x, y, z, w)).$$

Consider the level set $L = \{(x, y, z, w) : \mathbf{F}(x, y, z, w) = (0, 0)\}$. Suppose $\mathbf{a} = (a_1, a_2, a_3, a_4) \in L$ and

$$D\mathbf{F}(\mathbf{a}) = \begin{pmatrix} 2 & 3 & -2 & 4 \\ 3 & 2 & 1 & -2 \end{pmatrix}.$$

- a) Are both z, w locally differentiable functions of x, y near $\mathbf{a}$ on L ? Justify your answer. If yes, compute $\frac{\partial z}{\partial y}(\mathbf{a})$.
- b) Are both x, y locally differentiable functions of z, w near $\mathbf{a}$ on L ? Justify your answer. If yes, compute $\frac{\partial y}{\partial z}(\mathbf{a})$.

True or False?

P2Q1. Determine whether the following statements are True or False. No justification is necessary.

- a) $\{(x, y) \in \mathbb{R}^2 : xy < e\}$ is an open and path connected subset of $\mathbb{R}^2$.
- b) For any C^2 function $f : \mathbb{R}^n \rightarrow \mathbb{R}$, its Hessian matrix is symmetric at every point.
- c) If a C^1 function $f : \mathbb{R} \rightarrow \mathbb{R}$ has $f'(0) = 0$, then f has no local inverse near 0.
- d) For any simple closed curve $\gamma : [a, b] \rightarrow \mathbb{R}^2$, the region enclosed within γ is bounded.
- e) Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$. If the directional derivatives $D_u f(0, 0)$ exist for any unit vector $u \in \mathbb{R}^2$, then f is continuous at $(0, 0)$.